

CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 1

CANDIDATE
NAME

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CLASS

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CHEMISTRY

Paper 2

8872/02

TUESDAY 13 SEPTEMBER 2011

2 Hours

Candidates answer Section A on the Question Paper

Additional Materials: Answer Paper

Data Booklet

Graph Paper (1 sheet)

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

[PILOT FRIXION ERASABLE PENS ARE NOT ALLOWED]

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

SECTION A

Answer **all** questions.

SECTION B

Answer **two** questions on separate answer paper.

At the end of the examination, fasten all work securely together.

The number of marks is given in brackets [] at the end of each question or part of the question.

For Examiner's Use	
Paper 1	30
Section A	40
Section B	40
5 [20]	
6 [20]	
7 [20]	
TOTAL	110

This document consists of **15** printed pages and **1** blank page.

[Turn over]

Section A

Answer **all** questions in this section.

1. Chlorine is a powerful oxidant and is commonly used in bleaches and disinfectants. As a disinfectant, chlorine compounds are used in swimming pools to keep them clean and sanitary.

Chlorine is able to react with aqueous sodium hydroxide according to the following equation:

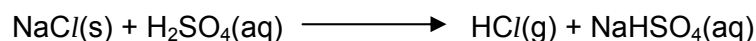


- (a) State the type of reaction that occurs when chlorine reacts with aqueous sodium hydroxide. Briefly explain your answer.

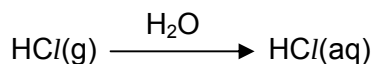
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..... [2]

- (b) In the above reaction, the sodium chloride formed can be used to produce hydrogen chloride gas by reacting sodium chloride with concentrated sulfuric acid, according to the following equation:



The hydrogen chloride gas is then bubbled into water, and subsequently diluted in a 250 cm³ standard flask to form hydrochloric acid, according to the following equation:



Calculate the volume of chlorine that needs to be reacted with aqueous sodium hydroxide at standard temperature and pressure to produce 0.2 mol dm⁻³ of hydrochloric acid.

- (c) 0.24 g of magnesium ribbon was then added to 100 cm³ of hydrochloric acid produced in (b) in a plastic beaker. The temperature at the start was 20.0 °C, and after the magnesium ribbon has dissolved, it rose to 30.0 °C.
[specific heat capacity of water = 4.18 J cm⁻³ K⁻¹].

- (i) Give the equation of the reaction between magnesium and hydrochloric acid.

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- (ii) Calculate the enthalpy change of the above reaction.

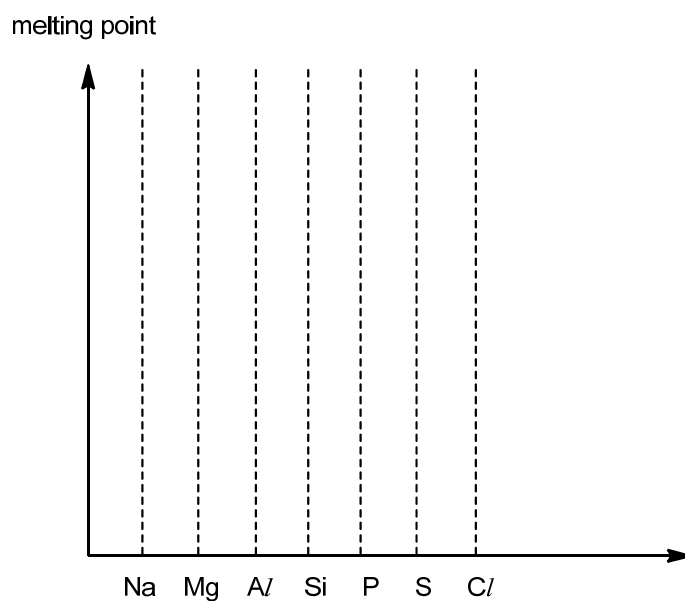
[4]

[Total: 9]

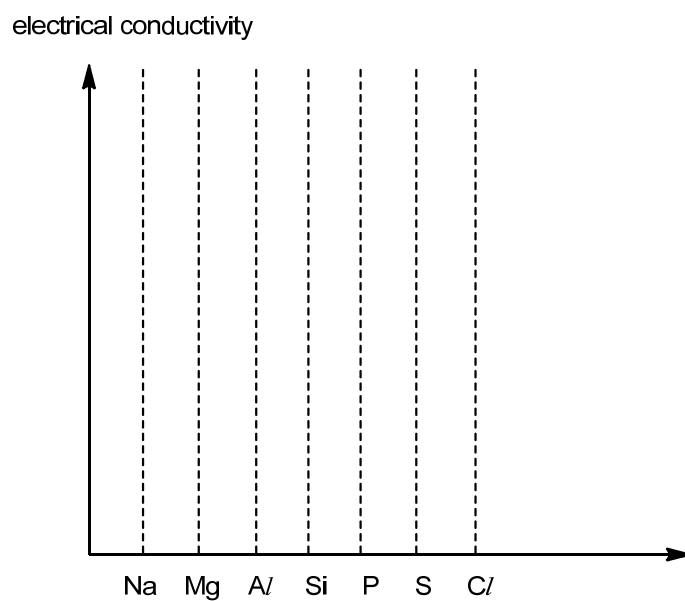
2. Magnesium and aluminium are Period 3 elements in the Periodic Table.

(a) On each of the graphs below, sketch the trends of the properties of the Period 3 elements across the Periodic table.

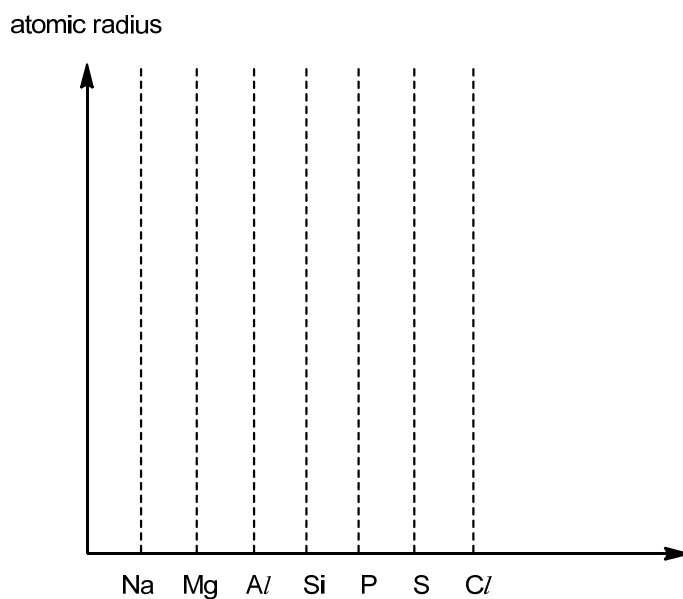
(i) melting point



(ii) electrical conductivity



(iii) atomic radius



[3]

- (b) Explain why aluminium has a lower first ionisation energy compared to magnesium.

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..... [2]

- (c) Suggest reasons for the following observations. Write balanced equations for any reactions that occur.

- (i) Magnesium oxide is slightly soluble in water but aluminium oxide is insoluble.

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- (ii) Both magnesium and aluminium chlorides can form acidic solutions in water, but aluminium chloride forms a more acidic solution than magnesium chloride.

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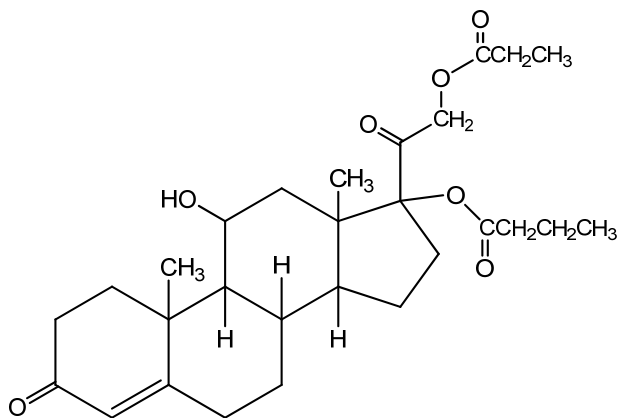
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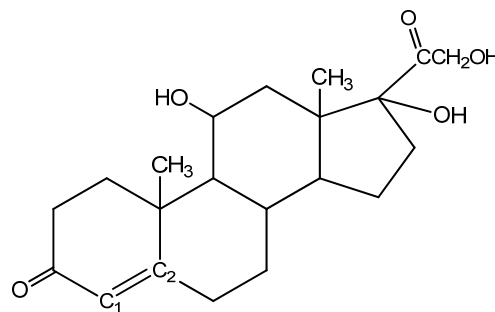
[6]

[Total: 11]

3. Hydrocortisone and cortisone are types of steroids which are applied to the skin to reduce inflammation, redness and swelling. Their structures are shown below:



Hydrocortisone



Cortisone

- (a) With reference to the functional groups present in the structures above, determine which of the two compounds is more soluble in water. Explain your answer.

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..... [2]

For parts (b) and (c), refer to cortisone only.

- (b) (i) Explain why geometric isomerism is **not** possible at carbons C₁ and C₂.

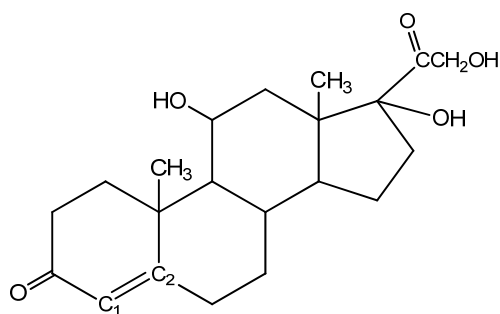
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- (ii) Draw the products formed after cortisone undergoes the following reactions:

I. heating with acidified K₂Cr₂O₇

For your convenience, the structure of cortisone is reproduced here.



Cortisone

II. adding acidified KMnO_4 at room temperature

[4]

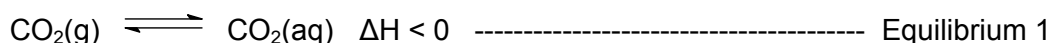
(c) (i) Which functional group(s) present in cortisone reacts with phosphorus(V) chloride?
Functional group(s):

(ii) Calculate the mass of PCl_5 required to react with 1 mg of **cortisone** ($M_r = 360$).

[4]

[Total: 10]

4. Carbonated drinks or 'fizzy drinks' (including beer) are best consumed when they are chilled. CO_2 is the gas dissolved in these drinks to give them the 'fizz'. In order for CO_2 to remain in the drink for long periods, the drinks should not be opened. This provides a closed system that allows equilibrium to be established between the CO_2 gas in the air above the drink and the CO_2 dissolved in the drink.

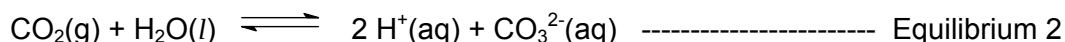


CO_2 is forced into 'fizzy' drinks under considerable pressure as high pressure enables more gas to dissolve. When the bottle of drink is opened, a 'pop' or 'fizz' sound is heard. If the drink is not consumed within a few hours after opening, the 'fizz' eventually disappears. At room temperature and pressure, only 90 cm^3 of CO_2 dissolves in every 100 cm^3 of water.

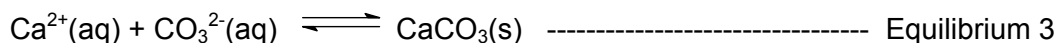
This process of CO_2 dissolving in aqueous solutions occurs everyday naturally, and on a much larger scale. CO_2 dissolves in the oceans such that the gas is taken out of the atmosphere to help maintain a stable environment on earth.

CO_2 is more soluble in water than O_2 or N_2 . Hence, much of the excess CO_2 released into the atmosphere from the combustion of fuels is absorbed by the oceans. The cold waters of the Arctic Ocean are expected to absorb more CO_2 than the warmer regions of the earth.

CO_2 molecules are able to react with water to form carbonic acid (H_2CO_3), and the simplified equation that represents this process is:



Near the surface of the oceans, many marine organisms build protective shells composed of insoluble CaCO_3 by using the CO_3^{2-} in the seawater.



This allows for more CO_2 to dissolve in the oceans. However, despite what is perceived as a greater amount of carbonic acid forming in the oceans, the oceans have been effective in controlling its pH. This is especially important for the oceans to support life. The ocean is thus, considered a buffer solution. The most common type of buffer is made up of a weak acid and one of its salts. The buffering action in the ocean is by carbonic acid, H_2CO_3 and hydrogen carbonate, HCO_3^- .

- (a) With reference to Equilibrium 1, use Le Chatelier's principle to explain why
- (i) If the drink is not consumed within a few hours after opening, the 'fizz' eventually disappears.
-
-
- (ii) The cold waters of the Arctic Ocean are expected to absorb more CO_2 than the warmer regions of the earth.
-
-

[4]

- (b) (i) At room temperature and pressure, calculate the amount, in moles, of CO_2 that can dissolve in 500 cm^3 of water.

- (ii) At room temperature, an unopened bottle containing 500 ml of carbonated drink has 0.300 moles of CO_2 dissolved in it. How much greater is the pressure in the drink compared to atmospheric pressure?

(Pressure and number of moles of gas are directly proportional.)

[2]

- (c) (i) Define the term 'buffer solution'.

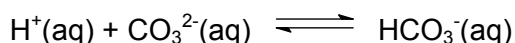
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- (ii) Using a relevant equation, show how the buffering system in the ocean works when it gets too alkaline.

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[2]

- (d) If the ocean gets too acidic, the following reaction will occur:



Using the information provided above, suggest and explain what would happen to the protective shells built by marine organisms if the oceans get too acidic.

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[2]

[Total: 10]

Section B

Answer **two** of the three questions in this section on separate answer papers.

5. Azo-compounds are compounds that contain the $R-N=N-R'$ (where R and R' are alkyl groups) functional group. Azo-compounds are commonly used as dyes because they have vivid colours like red, yellow and orange. Some of these compounds can also be used as indicators in acid-base titrations. The simplest azo-compound is diazomethane, $CH_3N=NCH_3$.

- (a) (i) Draw the structure of diazomethane, showing clearly the shape of the molecule. State the bond angles around the carbon and nitrogen atoms.
- (ii) With reference to its structure and bonding, comment on
- I. the boiling point; and
 - II. the solubility of diazomethane in water.

[8]

- (b) When diazomethane was heated, it decomposed to form ethane and nitrogen.

- (i) Construct an equation for the decomposition of diazomethane.
- (ii) Using relevant bond energy values in the *Data Booklet*, calculate the enthalpy change for the above decomposition reaction.
- (iii) Suggest a reason why the **method** in (b)(ii) cannot be used to calculate the enthalpy change of formation of sodium chloride.

The *partial pressure* of a gas is the pressure that the gas would exert if it occupies the container alone, and can be taken to be equivalent to its concentration.

The decomposition reaction was followed experimentally by measuring the *partial pressures* of diazomethane at various times t and the following results were obtained at 600 K.

t/s	partial pressure of azomethane/Pa
0	10.8
1000	7.4
2000	5.0
3000	3.4
4000	2.4

- (iv) By plotting a graph of partial pressure against time, show that this decomposition is a first order reaction. Hence, write the rate equation for the reaction.
- (v) Calculate a value of the rate constant for the reaction.
- (vi) Use your graph to determine the partial pressure of diazomethane at 1500 s.
- (vii) If the initial pressure of diazomethane is increased, explain, using *Collision Theory*, the effect on the initial rate of reaction.

[12]

[Total: 20]

6. Food digestion involves a chemical process called *hydrolysis*. Proteins, carbohydrates and fats are the three main food chemical groups that can undergo *hydrolysis*.

(a) (i) Explain the term *hydrolysis*.

(ii) Using a suitable organic compound of your choice, write a balanced chemical equation to show a *hydrolysis* process. State the reagents and conditions clearly.

[3]

Fat is an excellent storage medium with an energy content of 9 Kcal g^{-1} (Kcal g^{-1} is kilocalories per gram). This is more than the energy released from carbohydrates (eg. lactose), which is about 4 Kcal g^{-1} . These values represent the amount of energy released during combustion.

(b) (i) Define the standard enthalpy change of combustion.

(ii) Write a balanced equation, with state symbols, for the enthalpy change of combustion of lactose, $\text{C}_{12}\text{H}_{22}\text{O}_{11}$.

(iii) Show that the standard enthalpy change of combustion of lactose is $-5746 \text{ kJ mol}^{-1}$.

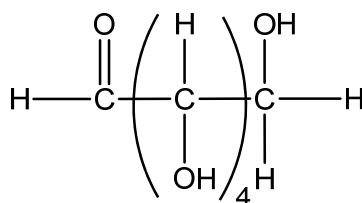
(Given: $1 \text{ Kcal} = 4.2 \text{ kJ}$ and the M_r of lactose is 342)

(iv) Hence, find the standard enthalpy change of formation of lactose.

(Given: standard enthalpy change of formation of $\text{CO}_2(\text{g}) = -393 \text{ kJ mol}^{-1}$, standard enthalpy change of formation of $\text{H}_2\text{O}(\text{l}) = -268 \text{ kJ mol}^{-1}$)

[6]

People who have lactose intolerance are unable to digest milk and milk products. These people lack an enzyme called lactase, which is responsible for hydrolysing lactose. Galactose is one of the products of the hydrolysis of lactose. The structure of galactose is shown below:



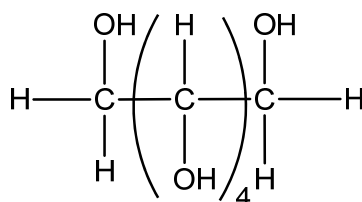
Galactose

(c) (i) Name the functional groups present in galactose.

(ii) State the catalyst and draw the organic product formed when **galactose** reacts with aqueous hydrogen cyanide.

[4]

Galactosemia, a genetic disease, is a result of absence of an enzyme that converts **galactose** into glucose. In a person with **Galactosemia**, a toxic compound called **dulcitol** forms instead. Although not fatal, **dulcitol** does cause severe stomach upset. The structure of **dulcitol** is shown below:



Dulcitol

- (d) (i) Suggest reagents and conditions that converts **galactose** into **dulcitol** in the laboratory. Name the type of reaction occurring.
- (ii) Suggest a simple chemical test to distinguish between **galactose** and **dulcitol**. [4]
- (e) Fatty acids are classed as saturated, monounsaturated or polyunsaturated, depending on the number of double bonds they contain. The iodine number, expressed in grams of iodine per 100 g of substance, is the amount of iodine that could add to the double bonds. Polyunsaturated fatty acids have two or three double bonds and have an iodine number of 181 and 273 respectively. State and explain the iodine numbers of
- (i) saturated fatty acids
- (ii) monounsaturated fatty acids

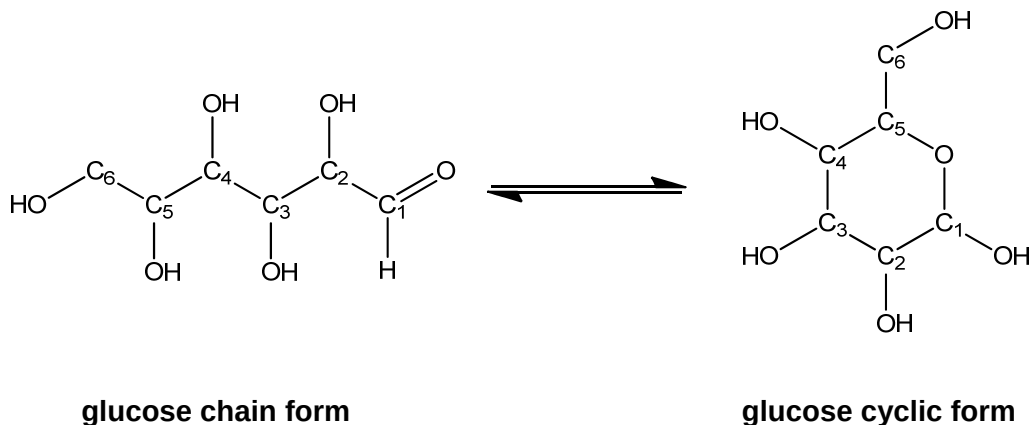
[3]

[Total: 20]

7. Glucose is referred to as a “reducing sugar” and it is an important carbohydrate in Biology. Cells use it as the primary source of energy.

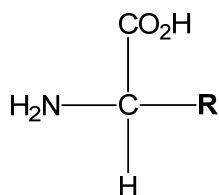
Glucose exists in different forms, either in chain or cyclic forms, as shown below. The straight-chain form is easily converted into the cyclic form when the -OH on C₅ binds to the C₁.

[Note: Some hydrogen atoms have been omitted to allow clarity of the structures.]



- (a) (i) State the type of reaction involved in the conversion above.
- (ii) Given that K_c for the conversion reaction is 5000, state the dominant form of glucose. Hence, find the percentage composition of each form at equilibrium.
- (iii) In its chain form, glucose reduces Benedict's solution (Fehling's). By considering the change in oxidation number, identify the species that has been reduced. State the functional group in glucose that allows it to be referred to as a “reducing sugar”. [6]
- (b) For diabetics who are unable to consume too much sugar, alternative sweeteners are needed. Aspartame is an artificial, non-saccharide sweetener used as a sugar substitute in some foods and beverages.

Hydrolysis of aspartame yields compound **A** shown below:



Compound A

[Note: **R** represents the rest of the molecule.]

The -NH₂ group is considered to be a base, a proton acceptor, due to the availability of a lone pair of electron on N atom. As such, compound **A** can react and dissolve in dilute aqueous HCl at room temperature.

- (i) Write an equation for the reaction of compound **A** with dilute aqueous HCl, and state the type(s) of interactions formed from this reaction.
- (ii) Compound **A** has the following composition by mass: 36.1% carbon, 5.3% hydrogen, 48.1% oxygen, 10.5% nitrogen.

Find the molecular formula of component **R** in compound **A**.

- (iii) Compound **A** reacts with 2 moles of Na_2CO_3 to form **B**. **A** also reacts with ethanol in the presence of concentrated sulfuric acid and heat to form **C**. When subjected under uv light and $\text{Cl}_2(\text{g})$, **A** forms a monochlorinated compound, **D**. When **D** reacts with alcoholic NaOH and heated, isomeric compounds **E** and **F** are formed.

Identify the structural formulae of compounds **A** – **F**, explaining the chemistry involved and writing equations where applicable.

[14]

[Total: 20]